

Create python flask app

```
priyansh@IDEAPAD-18UT:~$ mkdir ~/docker-webapp && cd ~/docker-webapp
```

Create 3 files

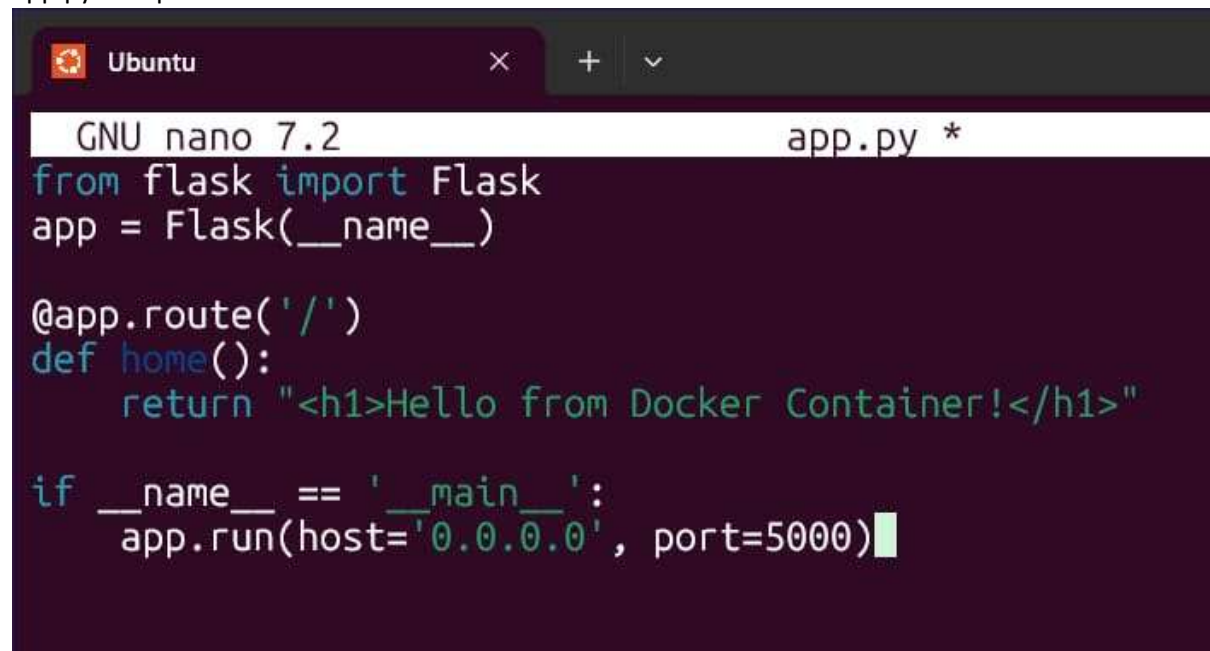
app.py – with python script

requirements.txt – 'flask'

Dockerfile – with python script

```
priyansh@IDEAPAD-18UT:~/docker-webapp$ nano app.py
priyansh@IDEAPAD-18UT:~/docker-webapp$ nano requirements.txt
priyansh@IDEAPAD-18UT:~/docker-webapp$ nano requirements.txt
priyansh@IDEAPAD-18UT:~/docker-webapp$ nano Dockerfile
```

app.py - script



```
GNU nano 7.2 app.py *
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "<h1>Hello from Docker Container!</h1>"

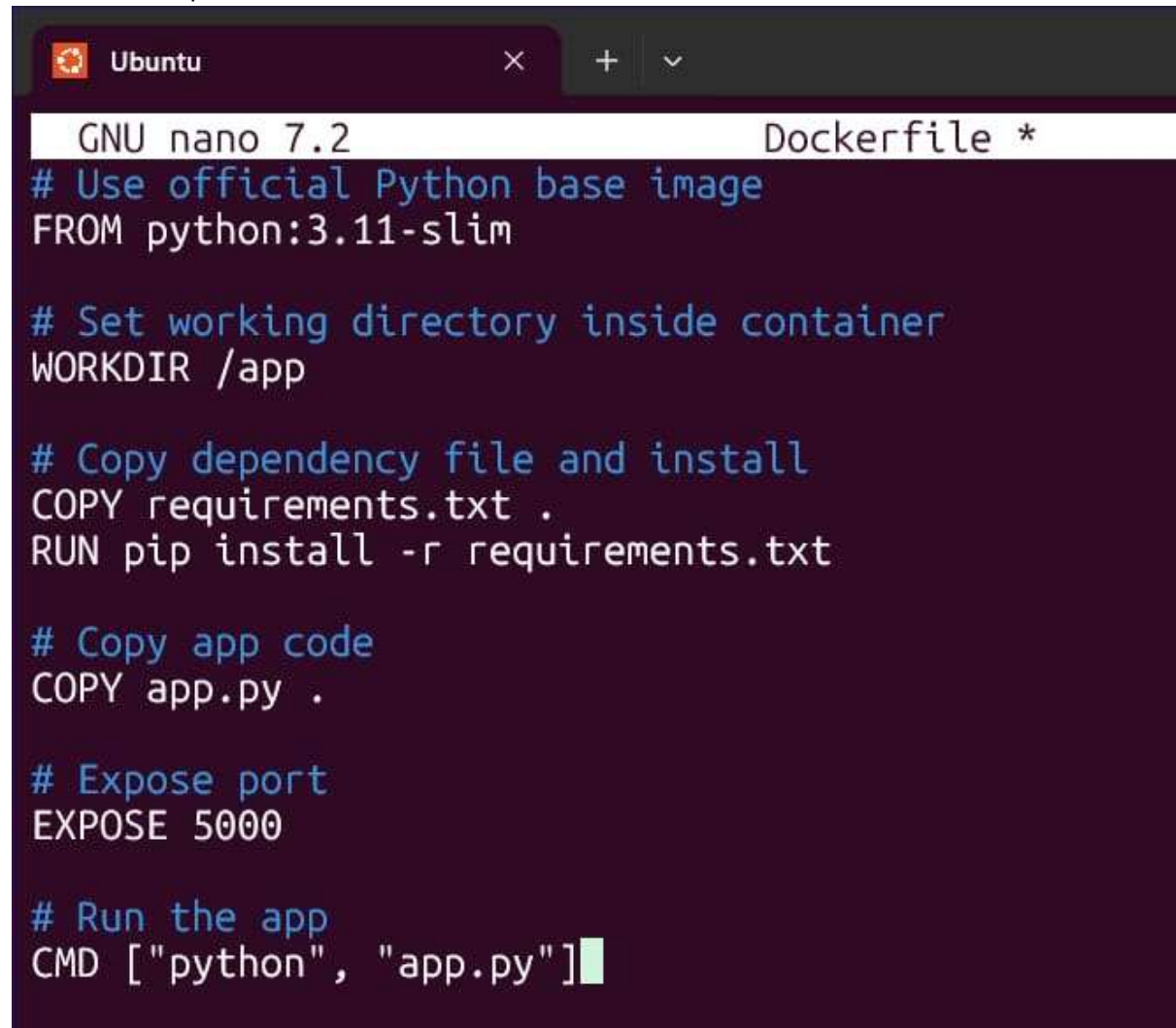
if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

requirements.txt



```
GNU nano 7.2 requirements.txt
flask
```

Dockerfile - script



The image shows a terminal window with a dark background. At the top, there is a title bar with the Ubuntu logo, the text 'Ubuntu', and window control buttons (close, maximize, and a dropdown arrow). Below the title bar, the terminal shows the GNU nano 7.2 editor interface. The file being edited is named 'Dockerfile *'. The content of the file is a Dockerfile script for a Python application. The script includes comments for each step: using the official Python base image, setting the working directory, copying and installing dependencies, copying the application code, exposing port 5000, and running the application. The script ends with a cursor on the last line.

```
GNU nano 7.2 Dockerfile *
# Use official Python base image
FROM python:3.11-slim

# Set working directory inside container
WORKDIR /app

# Copy dependency file and install
COPY requirements.txt .
RUN pip install -r requirements.txt

# Copy app code
COPY app.py .

# Expose port
EXPOSE 5000

# Run the app
CMD ["python", "app.py"]
```

Build, test and run the container

```
priyansh@IDEAPAD-18UT:~/docker-webapp$ # Build the image
docker build -t my-flask-app .

# Run the container
docker run -d -p 5000:5000 --name flask-container my-flask-app

# Test it
curl http://localhost:5000
# Or open browser → http://localhost:5000
[+] Building 69.6s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                 0.2s
=> => transferring dockerfile: 343B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-    7.5s
=> [internal] load .dockerignore                                   0.1s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d853    12.3s
=> => resolve docker.io/library/python:3.11-slim@sha256:6d8537    0.2s
=> => sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d 249B / 249B  0.7s
=> => sha256:f3ba2250c52436e41e9f2f3830af92d 14.37MB / 14.37MB  3.8s
=> => sha256:91ff8760033c3b9bc9c3a3bcc6ff66c11 1.29MB / 1.29MB  2.4s
=> => extracting sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e  1.9s
=> => extracting sha256:f3ba2250c52436e41e9f2f3830af92de466fca  7.0s
=> => extracting sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0  0.1s
=> [internal] load build context                                   0.2s
=> => transferring context: 279B                                    0.0s
=> [2/5] WORKDIR /app                                             0.9s
=> [3/5] COPY requirements.txt .                                  0.2s
=> [4/5] RUN pip install -r requirements.txt                     27.1s
=> [5/5] COPY app.py .                                           0.9s
=> exporting to image                                             17.8s
=> => exporting layers                                             11.8s
=> => exporting manifest sha256:0b20c68e85981970e77d64805d9a62  0.1s
=> => exporting config sha256:5955be142a99e35e95739e63a0b3d1da  0.1s
=> => exporting attestation manifest sha256:af4befb53e1bddb5f5    0.1s
=> => exporting manifest list sha256:f0d6e9860875365046622079a    0.1s
=> => naming to docker.io/library/my-flask-app:latest              0.0s
=> => unpacking to docker.io/library/my-flask-app:latest           5.2s
6effe9f32e2be736a23b9f54b34fd3d1856402dee35294e5af101d4c75f93b46
```

Check logs and then stop container

```
priyansh@IDEAPAD-18UT:~/docker-webapp$ docker logs flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
priyansh@IDEAPAD-18UT:~/docker-webapp$ docker stop flask-container
flask-container
priyansh@IDEAPAD-18UT:~/docker-webapp$
```